Anjez

Smart printing for businesses

AI Product Development and Monitoring

Project 1 Report

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Project repository

https://github.com/abdullah-nm/anjez.git

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Project overview

An integrated platform for brand identity, printing, and company knowledge

Anjez: smart printing and company knowledge

An Arabic platform combining brand management, occasion planning, print design, and answers grounded in company documents. Its knowledge assistant uses retrieval-augmented generation (RAG), with company boundaries derived from the authenticated user session.

Key capabilities in the updated product

- A connected monitoring dashboard at /monitor, backed by stored company-specific metrics.

- Request duration and outcome tracking, p95 latency, and separate indexing, retrieval, and answer-generation results.

- Transparent design-background classification: an AI-generated bitmap or a branded SVG template.

- Occasion templates and in-app reminders using Riyadh time, without fixing unconfirmed Hijri dates.

Verification summary

All 32 tests passed across seven files in approximately 6.17 seconds. TypeScript validation passed. Static analysis reported 11 warnings without a blocking error.

Interpret results within their scope

Software tests verify behavior; real answer accuracy requires separate evaluation.

01 Problem and solution

Users, value proposition, and the product journey

The problem

Marketing and administrative teams need occasion-based printed materials that preserve brand identity, as well as quick access to printing policies scattered across files. Mixing documents or logos between customers creates operational risk, making tenant isolation part of the solution.

User journey

1. Sign in with a demonstration account assigned an administrator or member role.

2. Manage the logo and colors, then select an occasion, product, and design content.

3. Preview and save a branded design, with an explicit statement of its actual background type.

4. Upload and index company knowledge, then ask questions with source references.

5. Open the monitoring dashboard as an administrator to inspect errors, speed, and workflow outcomes.

MVP scope

The product includes brand, studio, knowledge, calendar, notifications, orders, and monitoring pages. Demonstration accounts and company records are sample data, not evidence of actual use or partnerships. Sample knowledge is initialized when an embedding provider is configured.

Value to validate

The intended benefits are less manual policy lookup, consistent brand use, and easier service troubleshooting. Time savings and business impact have not yet been measured, so these remain product objectives.

02 Technology and architecture

Frontend, backend, external APIs, and databases

Languages and runtime

TypeScript is the primary frontend and backend language, with HTML and CSS for presentation and SQL for database schemas and queries. Node.js runs the server; npm manages packages and scripts. React and React Router implement the interface, while Express provides API routes.

Development and verification tools

Vite serves and builds the frontend; tsx runs TypeScript on the server. Vitest and Supertest verify behavior and HTTP routes, and Oxlint performs static analysis. Zod validates input, Multer manages uploads, Mammoth extracts DOCX text, and unpdf extracts PDF text.

APIs and external services

The interface calls internal Express APIs for knowledge, designs, occasions, and monitoring. The server supports Gemini REST for generation and embeddings; generateContent calls appear in the source. OpenAI-compatible text and embedding providers and Supabase Storage are optional integrations. Configuration alone does not prove that a service is enabled or a model is available to an account.

Databases and storage

SQLite is the local default: one database stores knowledge, and another stores platform data including request_metrics. PostgreSQL with pgvector and HNSW indexing is an alternative knowledge backend. Logos and files are stored locally within company boundaries; logo copies can be uploaded to Supabase when configured.

Trade-off: SQLite simplifies local demonstrations. PostgreSQL adds operational, migration, and vector-dimension requirements. No load test comparing the two backends has been performed.

03 Data and AI workflows

Knowledge sources, retrieval, and design composition

Data collection and preparation

Administrators upload TXT, MD, CSV, PDF, DOCX, or HTML files. Text is extracted and split into chunks of up to 900 characters by default, with 140-character overlap. Embeddings are stored with company and document identifiers and chunk order. Document status and indexing errors are recorded.

RAG pipeline

The server authenticates the session and derives company scope, embeds the question, and searches only that company's chunks. By default it retrieves up to six chunks before similarity filtering, then sends context to the language model. When context is insufficient, it explicitly says so. Sources are shown by filename and chunk number.

Design composition in this version

The server builds an SVG template using brand colors, the original logo, and text. When a logo and image provider are available and generation succeeds, it adds an AI bitmap background. Otherwise, it can save a branded template with an explicit explanation. background_kind distinguishes ai_bitmap_background from branded_svg.

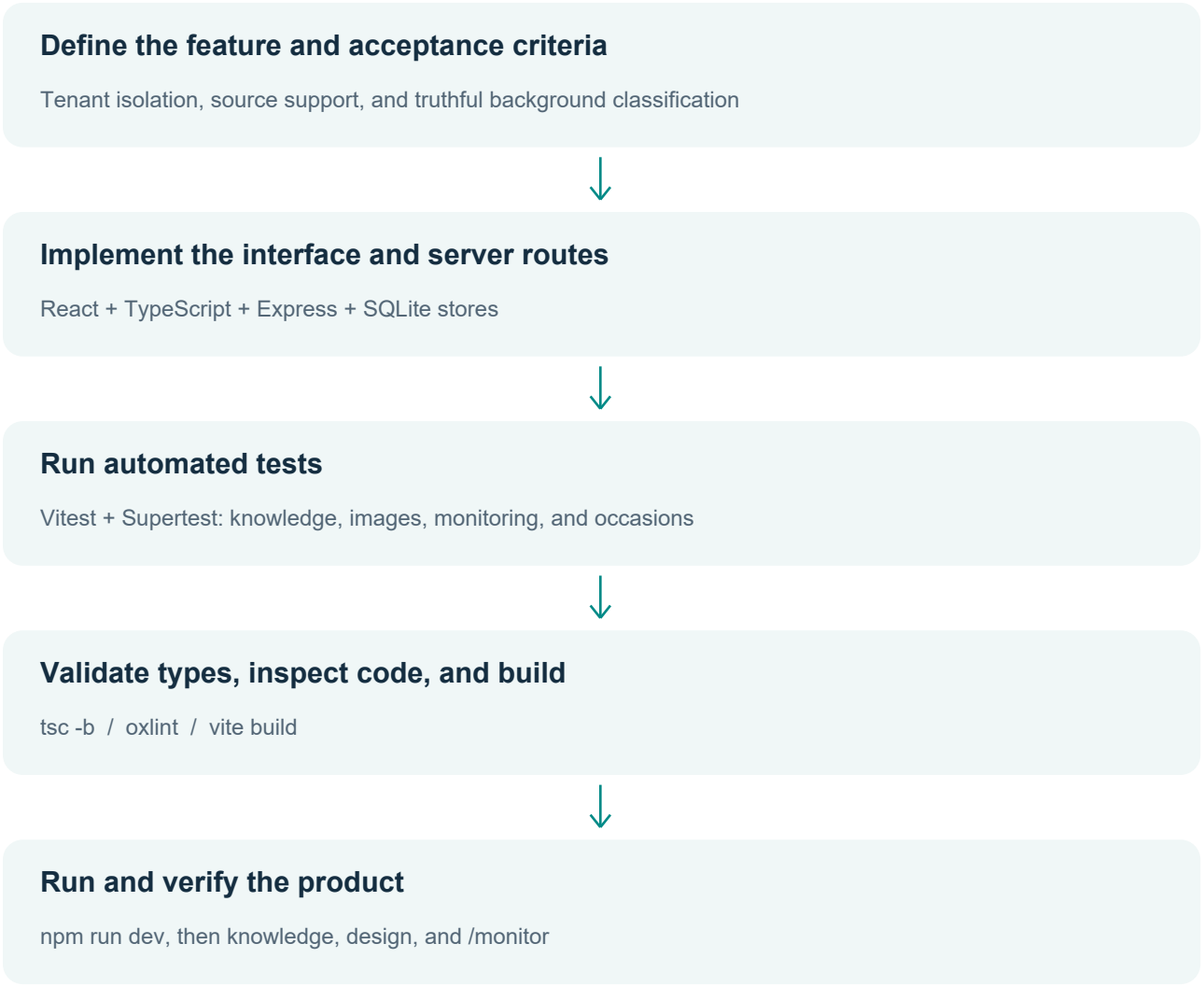
In compose.ts, design copy is currently assembled directly from the user's content, occasion, purpose, and audience. The presence of a Gemini client does not establish AI copy generation in this path.

Quality and protection

Client-supplied company_id is ignored; API keys remain on the server. Embedding dimensions are checked. The PostgreSQL vector column has 1,536 dimensions and must match the configured model. OCR for scanned files and a representative real-customer evaluation dataset have not been established.

04 Development workflow

Diagram 1 | Available tooling and the local delivery sequence



The diagram documents project tools and the local workflow; it does not establish automated CI/CD or production deployment. Failures are investigated through request outcomes, error codes, tests, and environment checks before verification is repeated.

Verification evidence

Verification includes route tests, type checking, and static analysis. Answer quality on real documents and successful live provider calls remain separate evaluation tasks.

05 Monitoring workflow

Diagram 2 | Actual data flow into the administrator dashboard



The summary endpoint is excluded from recording so dashboard reads do not inflate metrics. Requests without an attached authenticated company are omitted. Metric-write failures do not block product responses.

Privacy and access control

Unauthenticated access receives 401; members receive 403. Scope comes from the session. Metrics exclude question and answer text, document content, and keys. The dashboard displays classified errors; no automatic threshold-alert system was found in the reviewed source.

06 Monitoring metrics

Definitions, rationale, and limits of interpretation

Request count and error rate

Error rate is the number of recorded HTTP statuses >=400 divided by all records in the period. It includes client and server errors, not only 5xx. The rate is unavailable when there are no requests.

p95 response time

Durations are sorted and the value at rank ceil(0.95 x N) is selected. This exposes slow requests that an average may hide. The dashboard also shows busy routes. It measures recorded authenticated traffic, not all server activity.

Indexing, retrieval, and generation

The dashboard separates document preparation, search, and answer generation. A successful search with no chunks is not a model failure and does not call generation. Successful answer generation does not establish factual correctness.

Designs and provider errors

Preview, composition, and image requests are distinguished, as are bitmap backgrounds and SVG templates. Gemini error categories such as quota, auth, and unavailable support troubleshooting.

Health, tokens, and cost

Database health uses SELECT 1. Gemini health reflects configuration without a live probe. Tokens are counted when usageMetadata is available. Financial cost remains unavailable without an added pricing model.

Windows are one hour, 24 hours, seven days, and 14 days. Each is compared with an equal previous window, but 14-day retention limits historical completeness for long comparisons. An empty previous period does not prove there was no earlier activity.

07 Evaluation and results

Verified software behavior and a separate quality-evaluation plan

Verification results

All 32 tests passed across seven files in approximately 6.17 seconds. TypeScript validation passed without diagnostics. Static analysis reported 11 warnings and no blocking error, including effect-driven state updates, hook dependencies, export separation, and a control-character cleaning expression. Production performance and live provider calls were not evaluated.

Numerical evidence from the monitoring test

The test creates 100 synthetic requests with durations from 1 to 100 milliseconds and one failure. It verifies 100 requests, a 1% error rate, and a p95 of 95 milliseconds. These are controlled calculation checks, not real-user operating statistics.

Tests also verify company isolation, administrator-only monitoring, rejection of company-id manipulation, exclusion of question and document text from metrics, and continued service during metric-write failure. Occasion tests cover fixed Gregorian recurrence and undated unconfirmed occasions.

Proposed reference evaluation

Prepare 30 reference cases: 15 answerable from documents, 10 outside the knowledge base, and five company-boundary attacks. Define expected answers and sources, then have two reviewers assess correctness, source support, and appropriate abstention. Compare manual lookup and RAG completion time on the same questions.

This quality-evaluation plan has not been executed. Real answer accuracy, time savings, and financial cost remain unmeasured. No external model-quality benchmark was used. Mock-provider tests do not replace evaluating Gemini on representative documents.

08 Competitor benchmark

Market comparison | Design, printing, and business workflows

This descriptive comparison uses capabilities advertised on official websites. It includes Saudi printing providers and global design alternatives with overlapping use cases. It is not a price comparison, market-share ranking, or technical superiority test.

Competitor	Published capabilities	Anjez positioning opportunity
Canva	Brand Kit, AI tools, and marketing-material design and printing. [9, 10]	Combine company documents and occasion planning in one Arabic workflow; no claim of stronger editing tools.
VistaPrint	AI-assisted logo design and print-ready files. [11]	Use company knowledge and policies before design; commercial fulfillment is not established in the MVP.
Easy Print	Select products, design online or upload files, then pay and arrange delivery or collection. [12]	Create in the context of company identity and occasions; the competitor has an explicit fulfillment journey.
Zeej Print	Business and marketing print products, online ordering, and additional professional design services. [13]	Integrate brand, knowledge, and occasions; product variety and finishes are competitor strengths.
Print.sa	Upload PDFs, select print options and a nearby provider, and collect or receive delivery. [14]	Prepare branded content before ordering; a printer network is not a demonstrated Anjez capability.

Proposed market position

An Arabic assistant for preparing company print materials, combining brand identity, policy knowledge, and occasion planning. This positioning is an inference to validate with customers. RAG or occasion planning not being mentioned on a reviewed page does not establish their absence from every competitor service.

09 Project development tools

Tools used and their roles, as reported by the team

Working tools

Tool	Role in the project
Cursor	Development environment for writing code, with coding assistance to understand, modify, and review project files.
ChatGPT	Support for ideas and requirements, explanations of solutions, and content and documentation preparation.
GitHub	Source hosting and sharing, version management, and collaboration among team members.

These tools support the development workflow. React, Express, TypeScript, and Gemini are product technologies described in the architecture section. The team reviews and tests AI-assisted outputs before adopting them.

Project repository

https://github.com/abdullah-nm/anjez.git

10 Next steps and references

Submission readiness, team understanding, and evidence

Alignment with project requirements

Product and environment: functional source and passing tests; a configured live demonstration is still needed. Monitoring: a connected dashboard with stored metrics and access controls. Diagrams and report: included here. Source handoff: the archive and team-provided GitHub repository. Presentation deck: prepare separately for the final presentation.

Limitations and priorities

Prioritize a reference answer-quality dataset, static-analysis warnings, clean installation, and live-service testing. Account for retention limits in historical comparisons and add automatic alerts if required. Reminders run at startup and hourly while the process is active; an entire due day can be missed if it is offline. Print PDF export with bleed and CMYK, email, and browser notifications are not enabled.

Tools and team participation

Project work includes interface and server development, data preparation, testing, and documentation. Each member should understand the complete workflow and explain technical decisions. Team tools include Cursor, ChatGPT, and GitHub, with human review and testing of generated outputs.

Internal references

[1] Project 1 instructions.pdf - product, monitoring, evaluation, and delivery requirements.

[2] anjez-master (2).zip - source version reviewed.

[3] README.md and package.json - setup, dependencies, and declared limits.

[4] server/app.ts and server/observability/ - recording, aggregation, and authorization.

[5] src/pages/Monitor.tsx - metric display and time windows.

[6] server/rag.ts, server/lib/chunk.ts, and server/store.ts - knowledge pipeline.

[7] server/platform/compose.ts, reminders.ts, and occasions.ts - design and calendar.

[8] server/test/*.test.ts - verification evidence.

11 Market comparison sources

Official pages supporting the competitor capabilities in the benchmark

[9] Canva - Brand Kit

Brand assets, colors, and templates.

https://www.canva.com/pro/brand-kit/?price=free

[10] Canva - Marketing Materials

Marketing-material design and printing.

https://www.canva.com/print/marketing-materials/

[11] VistaPrint - AI Logo Design

AI-assisted logo design and files for printing.

https://www.vistaprint.com/logo-design

[12] Easy Print

Products, online design or file upload, payment, and delivery.

https://easyprint.com.sa/

[13] Zeej Print

Business printing, marketing materials, and additional design services.

https://zeejprint.com/ar-SA

[14] Print.sa

Ordering file printing from local service providers.

https://www.print.sa/

The comparison reflects public pages reviewed during preparation. Services vary by country, plan, and product availability. It is not a company ranking or proof of absolute superiority.

Brand management

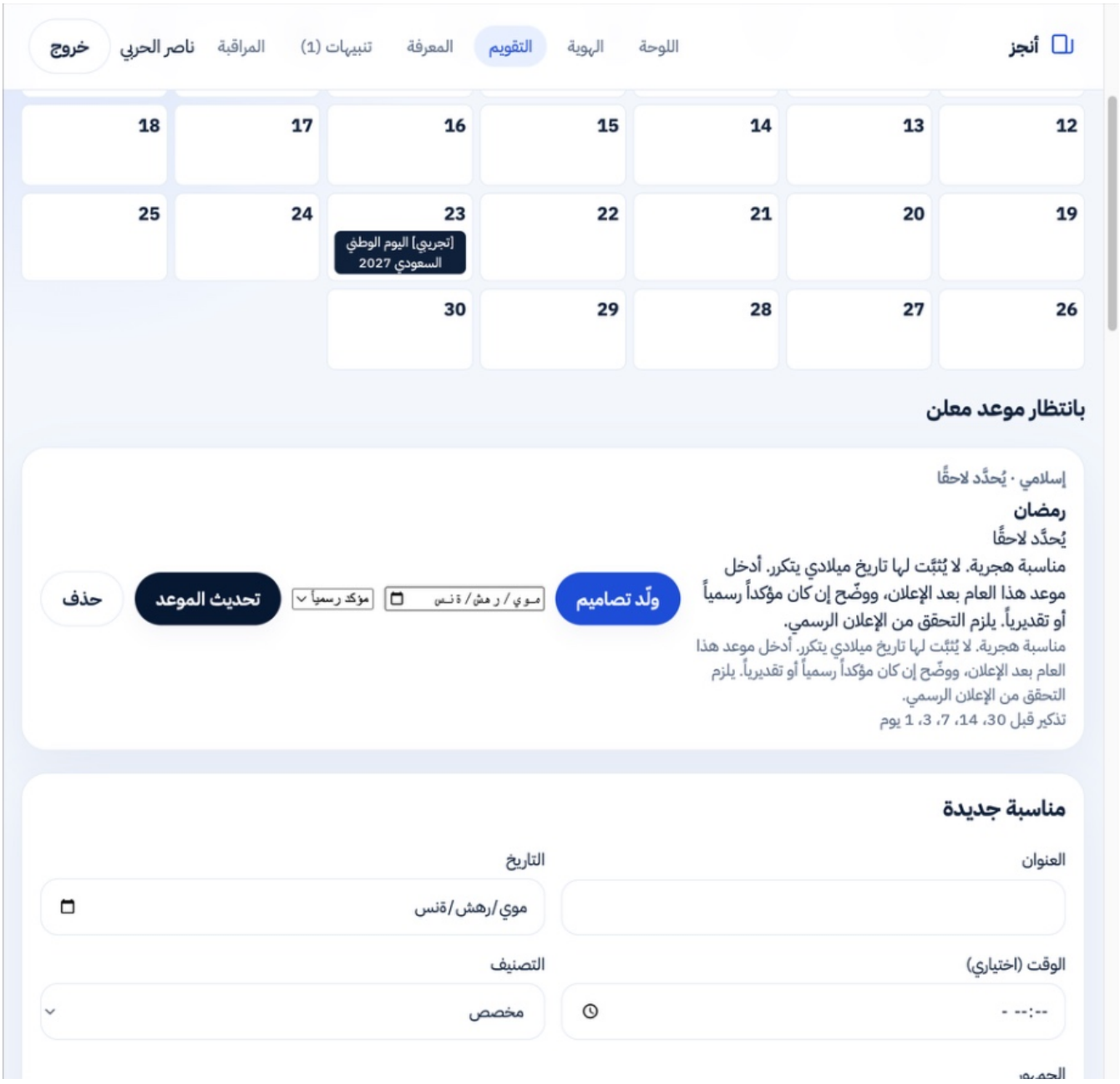
Project screenshots | 01 / 20



Company information, logo, colors, and content language.

Occasion calendar

Project screenshots | 02 / 20



Calendar dates connected to the design context.

Product selection

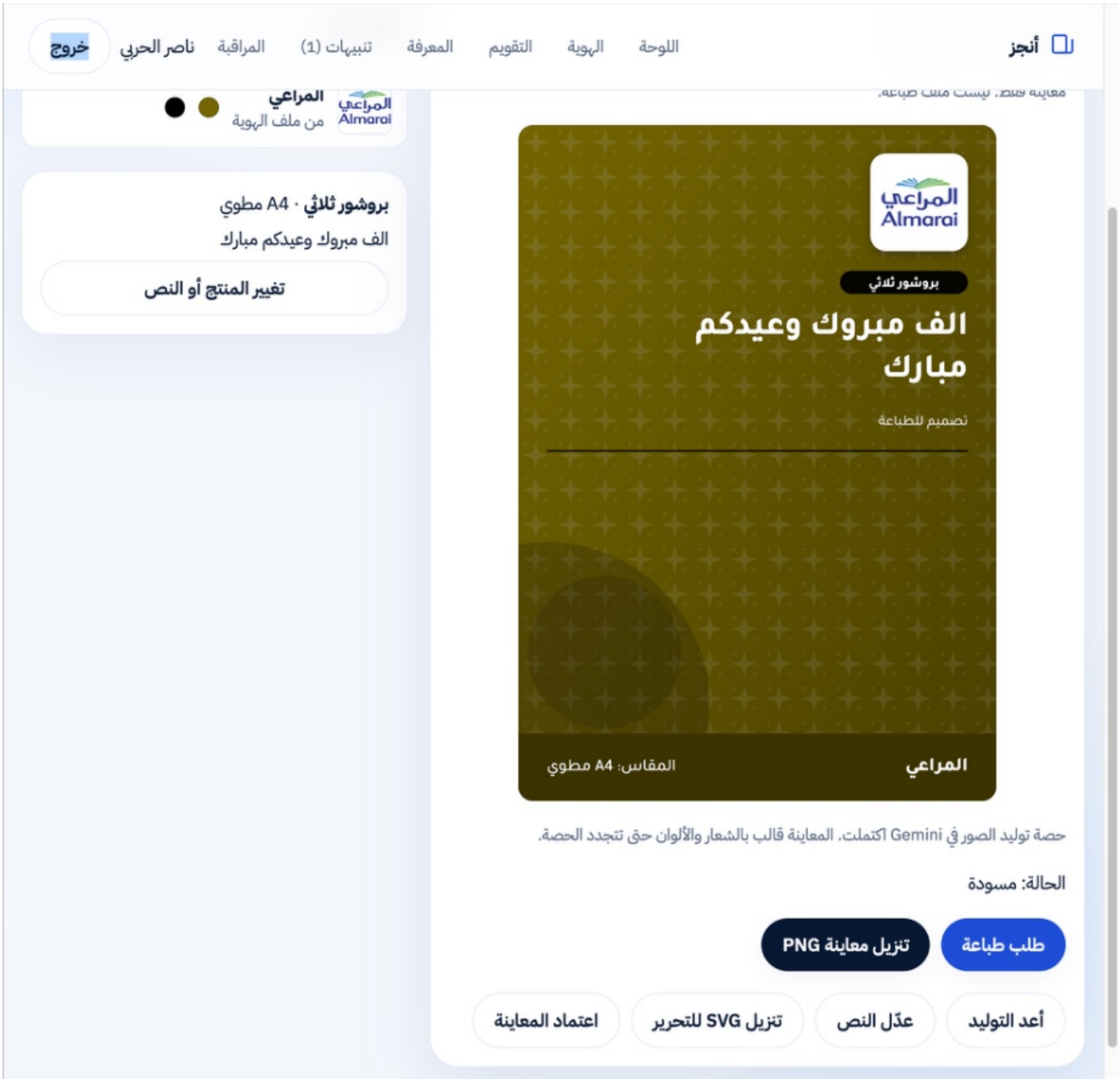
Project screenshots | 03 / 20



Print-product selection alongside a branded preview.

Design preview

Project screenshots | 04 / 20



Reviewing content and the logo before continuing.

Occasion template preview

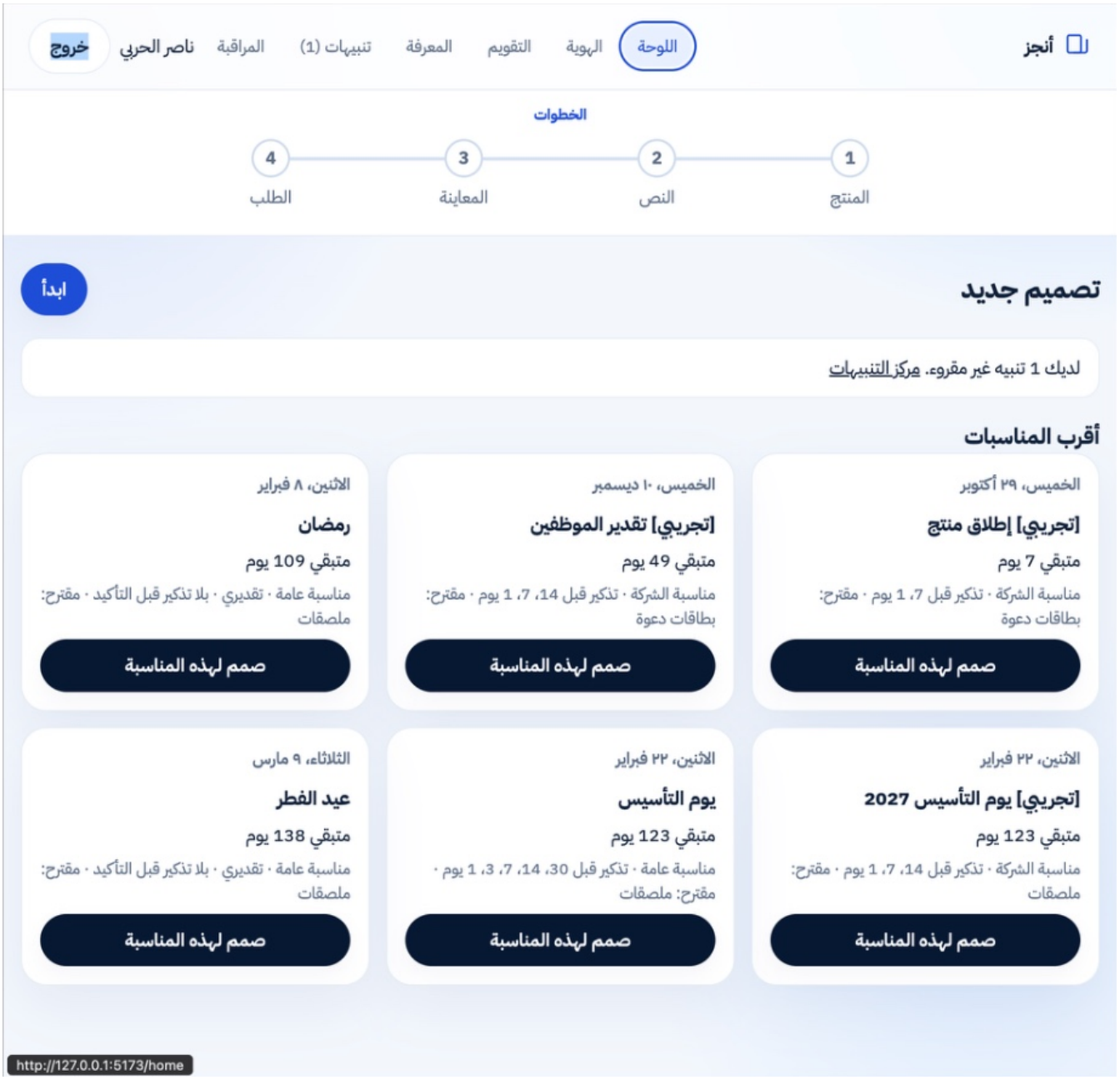
Project screenshots | 05 / 20



Comparing the selected product with its visual preview.

Home dashboard

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Browsing occasions and starting a new design.

Design studio

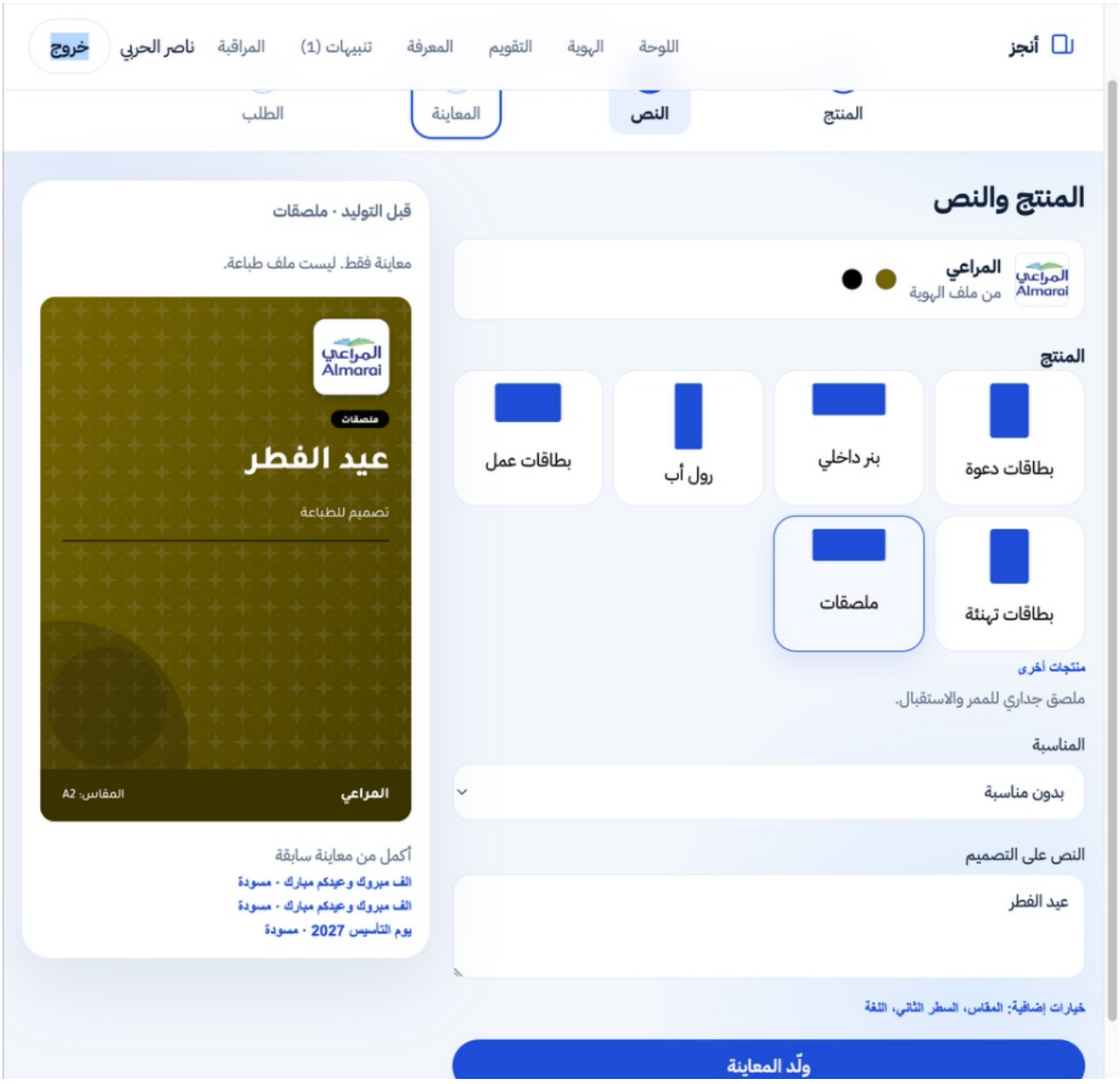
Project screenshots | 07 / 20



Selecting a product within the guided design workflow.

Design properties

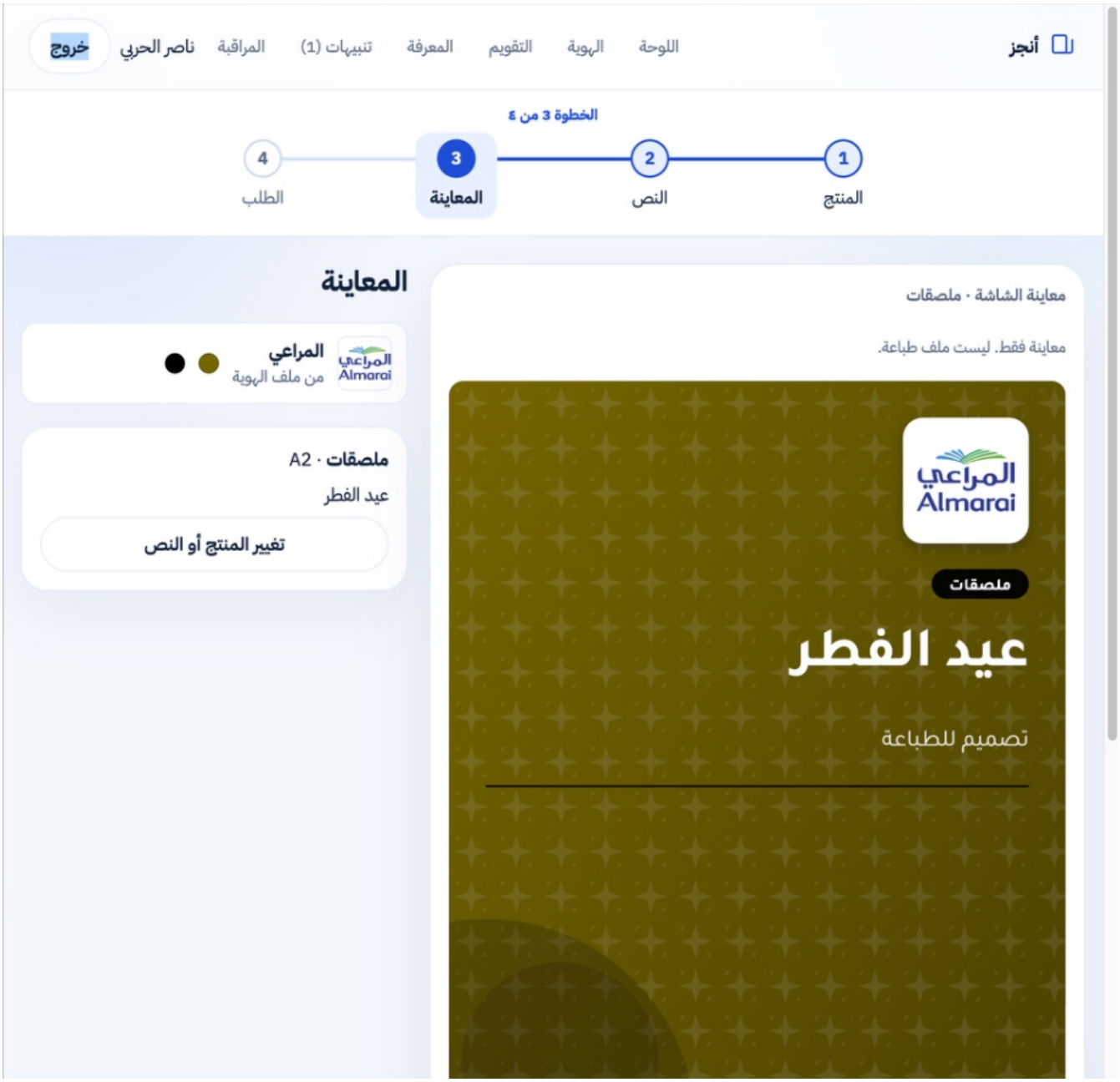
Project screenshots | 08 / 20



Adjusting options while reviewing the preview.

Preview verification

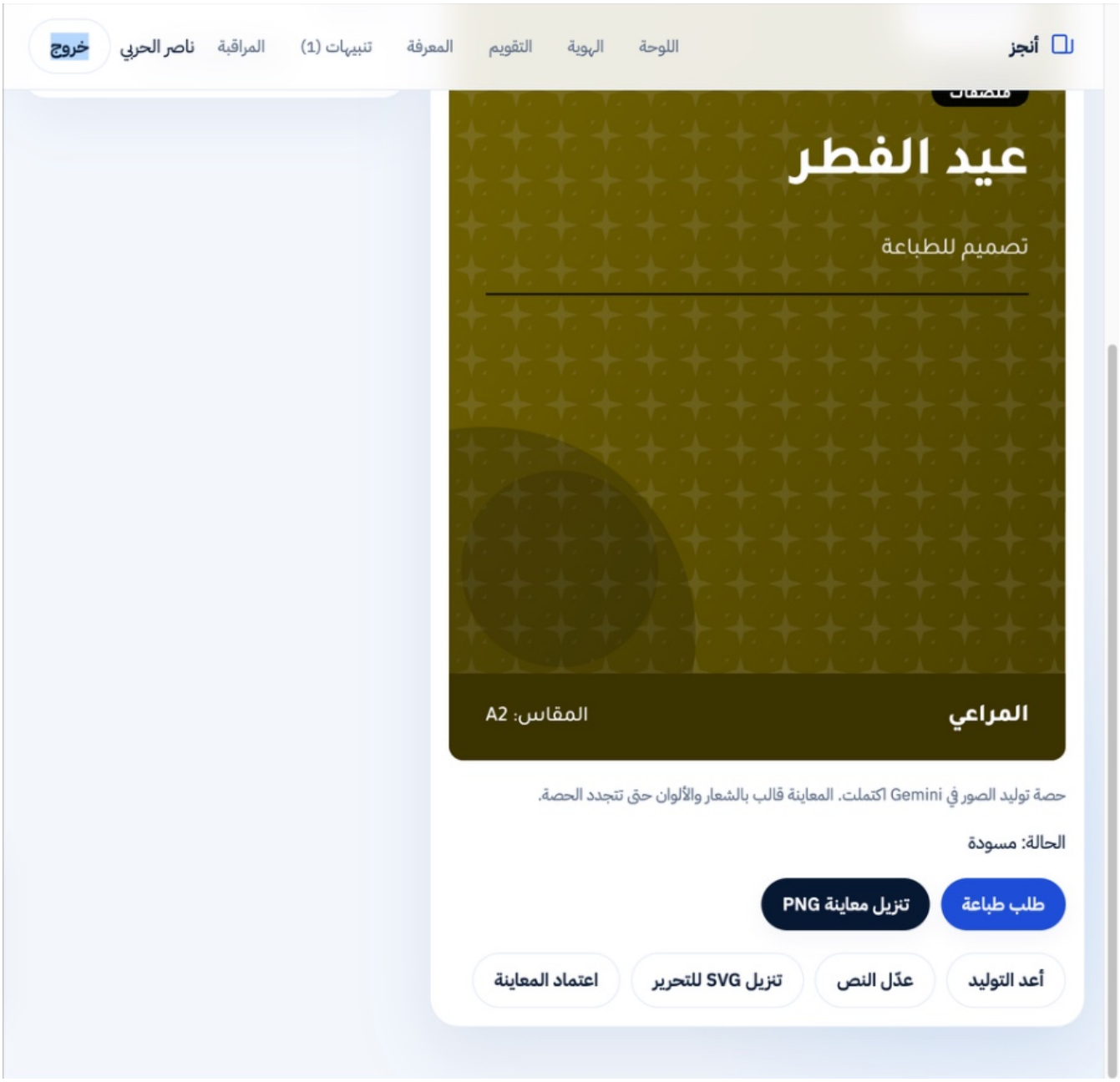
Project screenshots | 09 / 20



Checking size, brand identity, and template content.

Output details

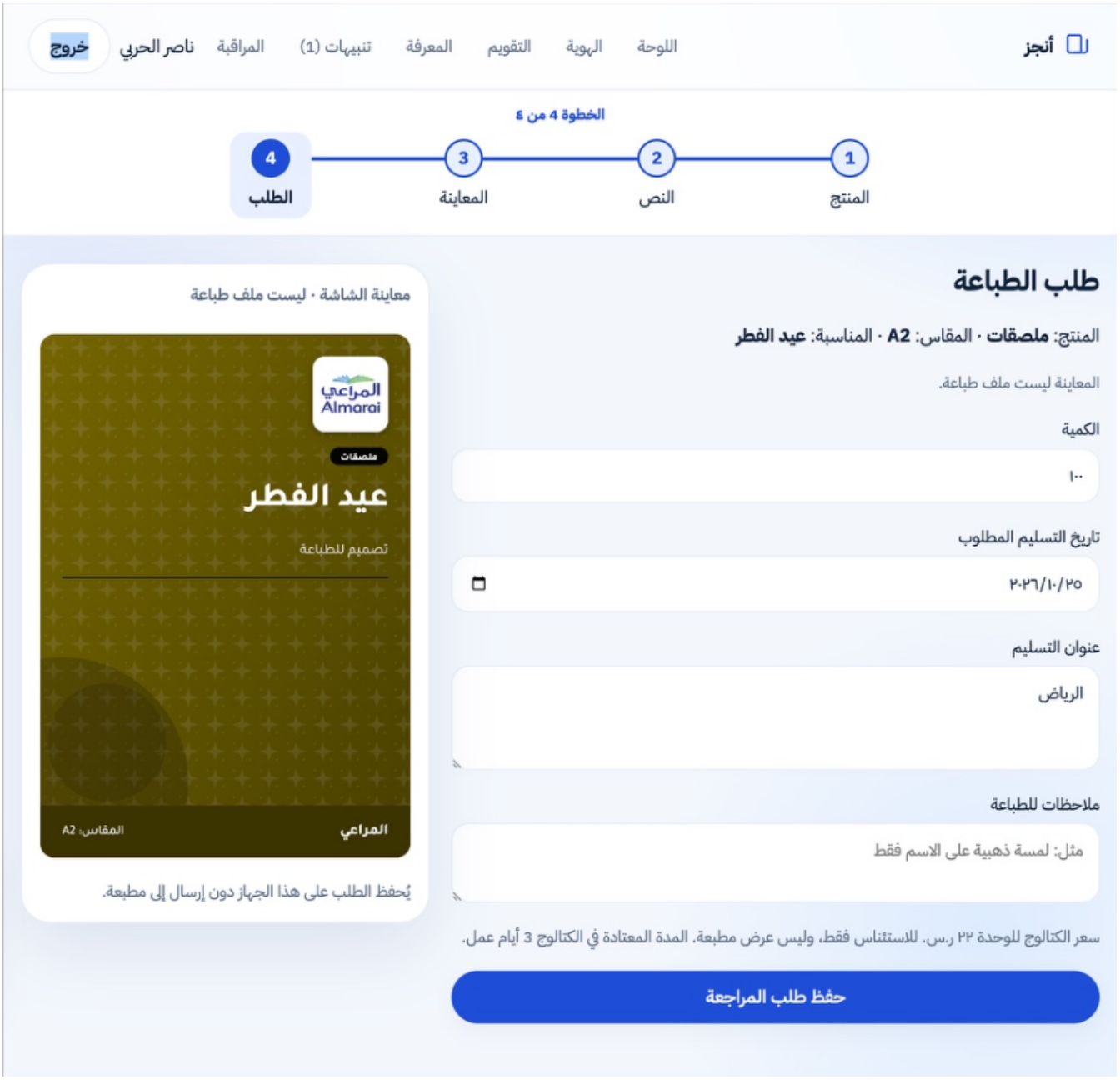
Project screenshots | 10 / 20



Inspecting text and the resulting design.

Print order

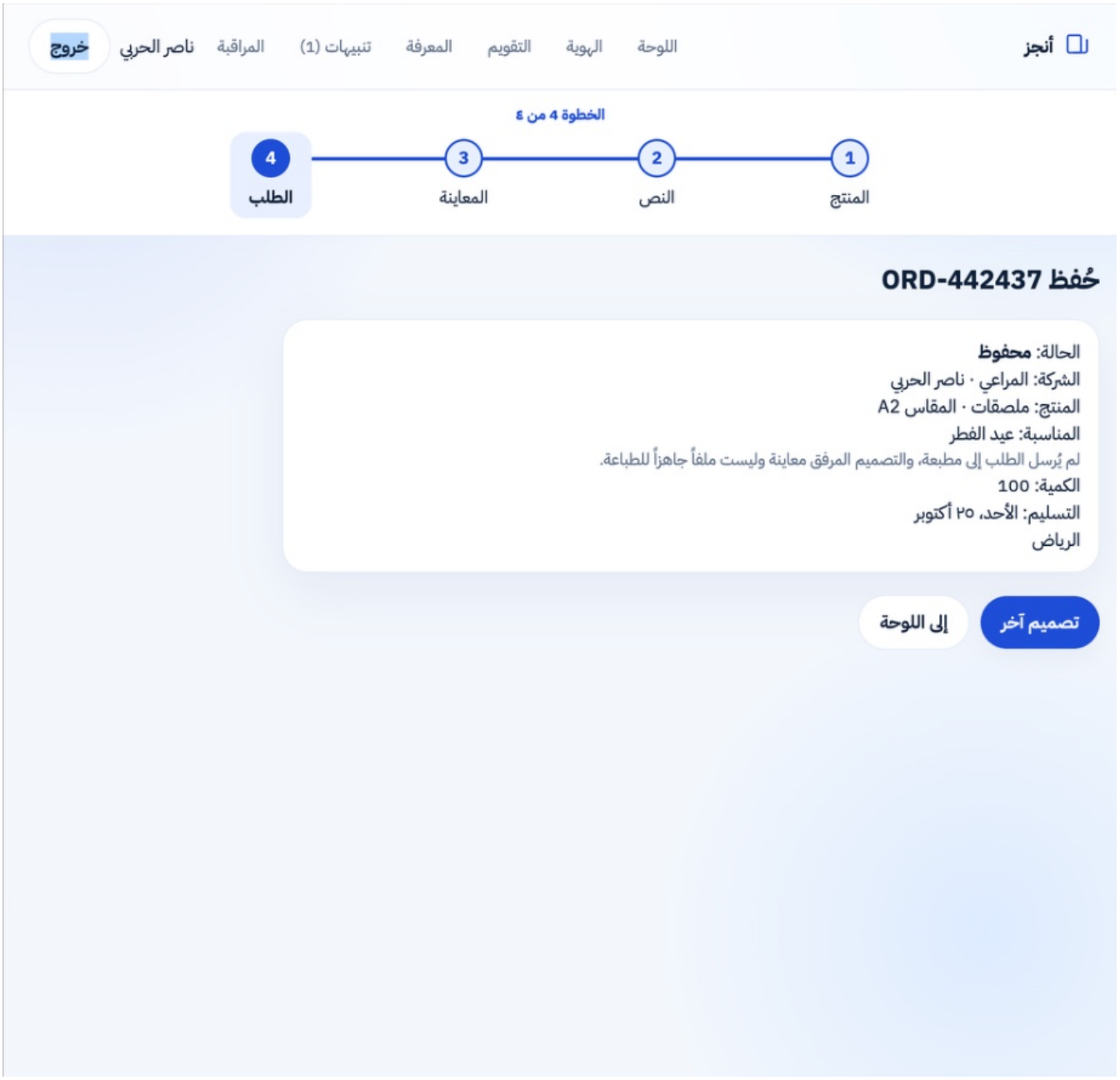
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Entering order details beside the print preview.

Order confirmation

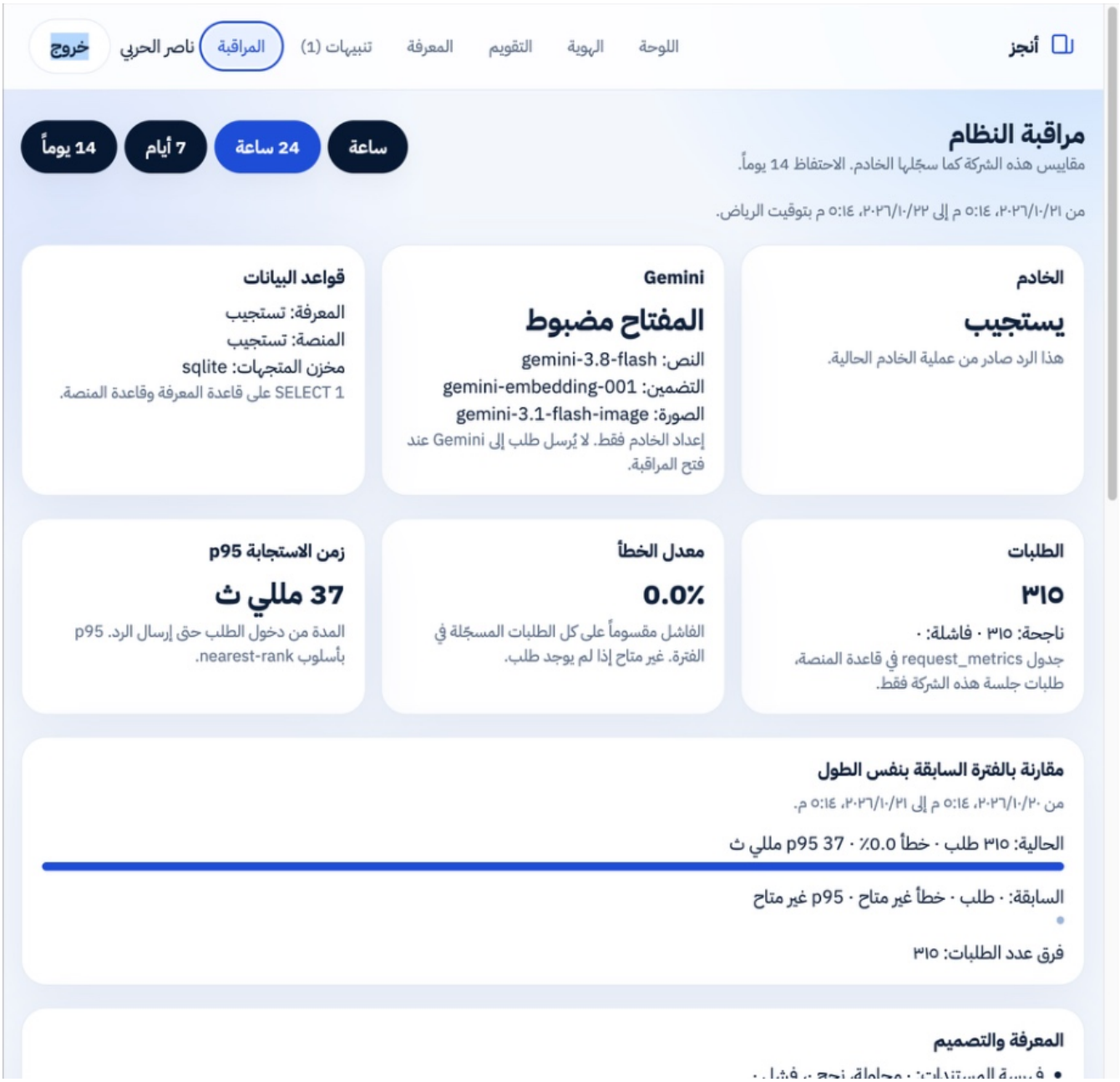
Project screenshots | 12 / 20



Displaying the order summary after submission within the platform.

System monitoring

Project screenshots | 13 / 20



Service health, request counts, error rate, and response time.

Service route monitoring

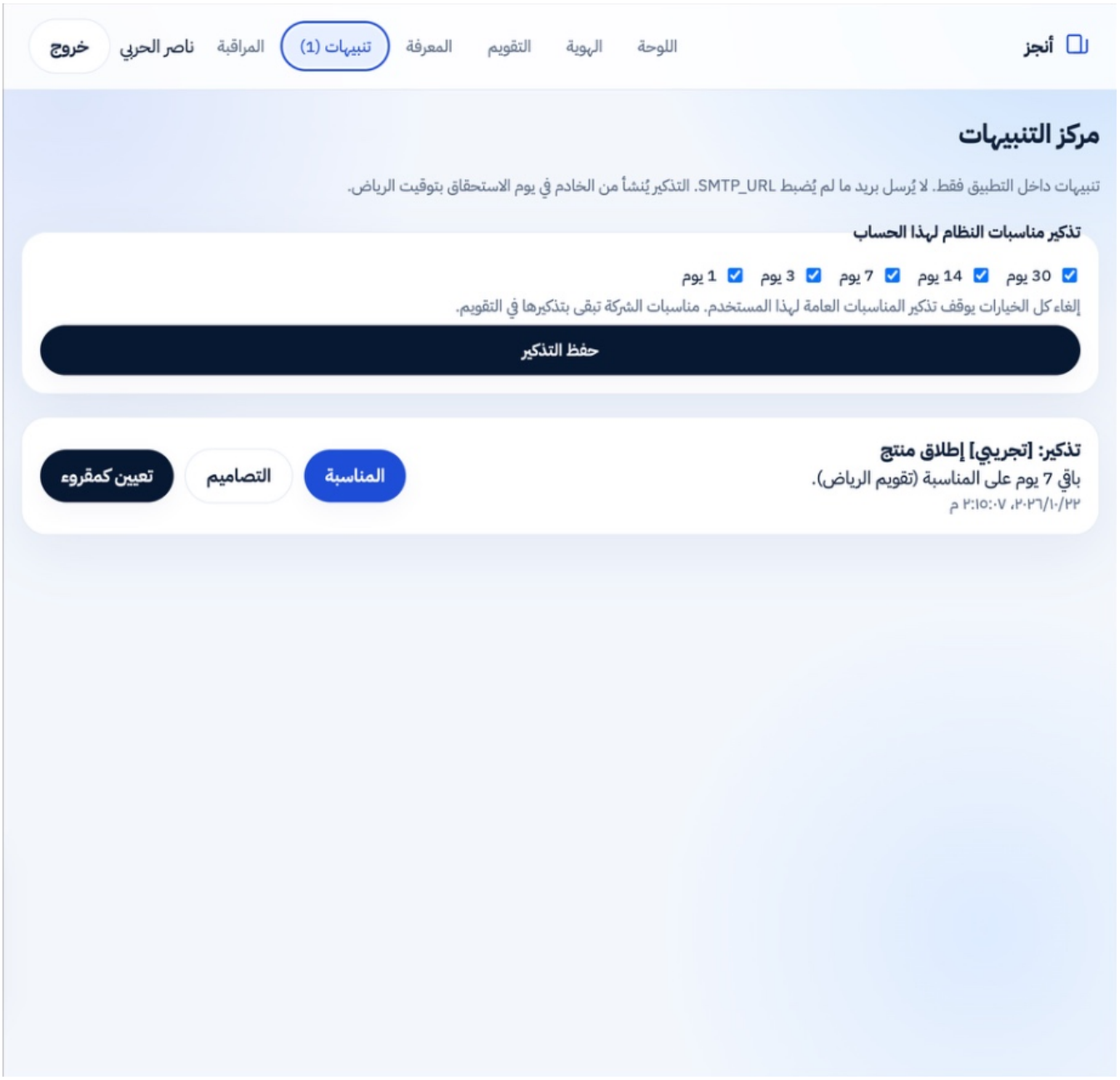
Project screenshots | 14 / 20



Successful and failed requests and p95 for individual routes.

Notification center

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Notifications associated with occasions.

Occasion details

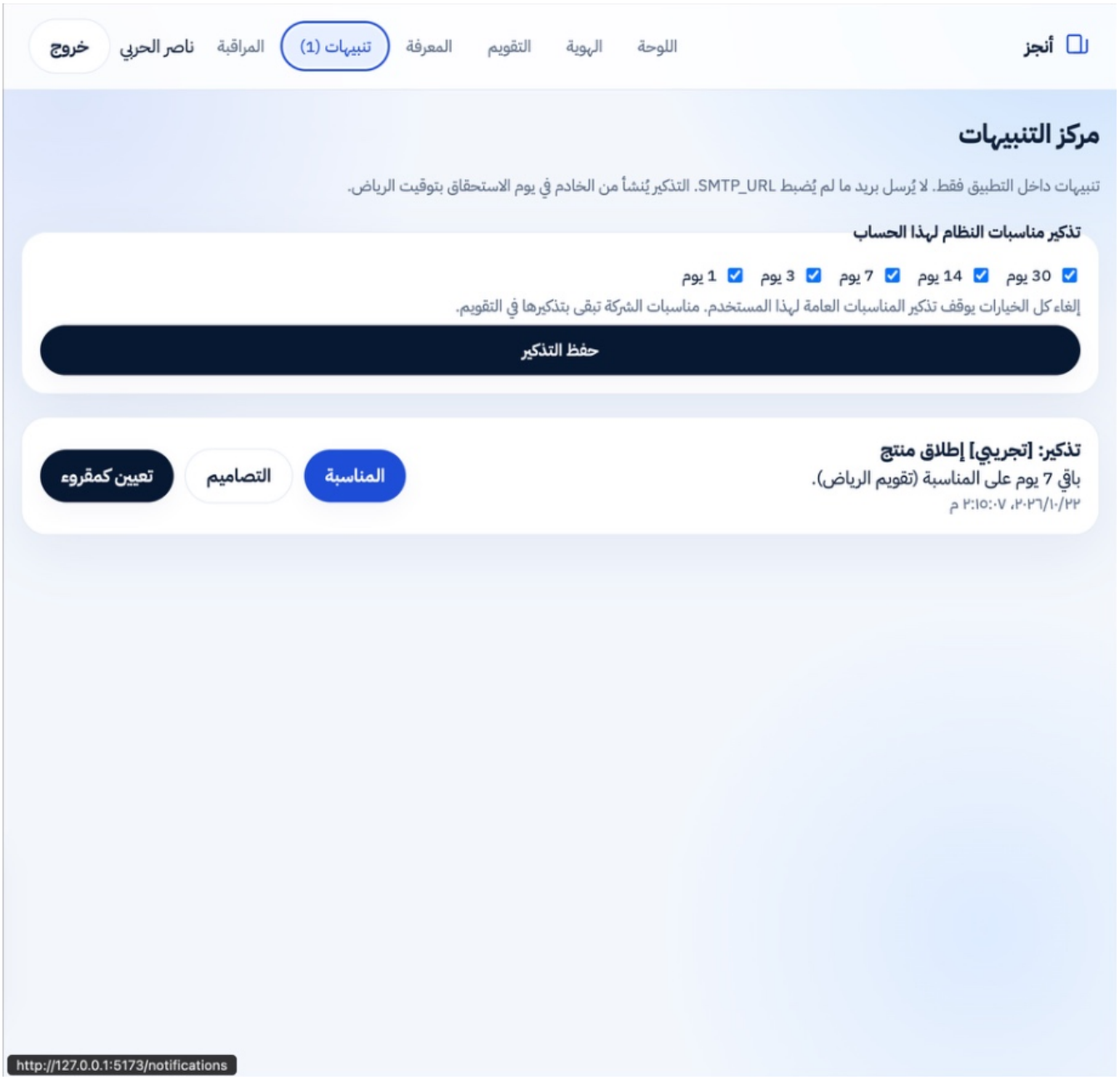
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Reviewing occasion records and their settings.

Notification list

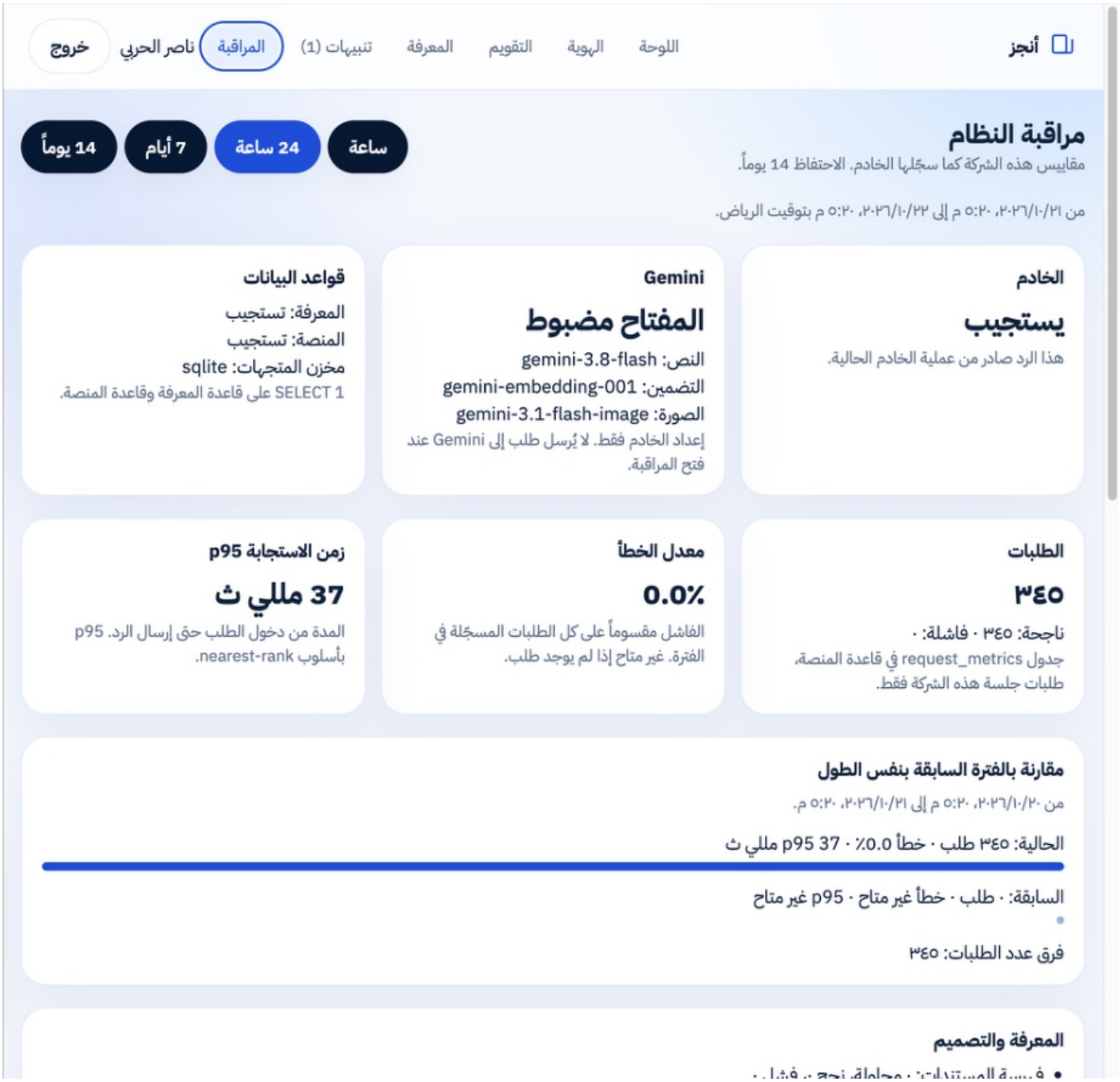
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Following in-app messages.

Operating indicators

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Monitoring summary for a selected time window.

Monitoring details

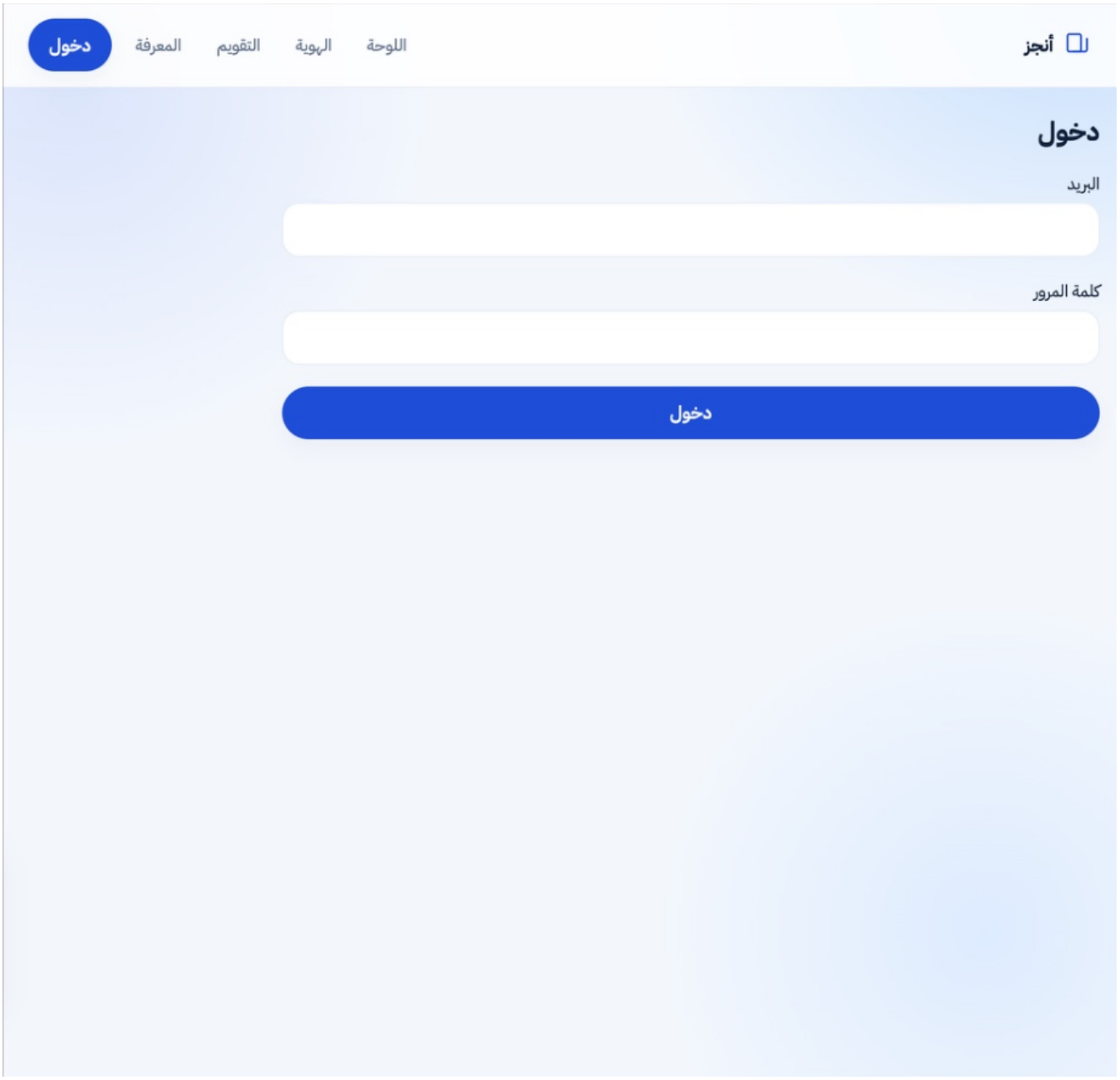
Project screenshots | 19 / 20



Provider information, routes, and the latest recorded errors.

Sign in

Project screenshots | 20 / 20



Accessing the company account.